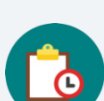
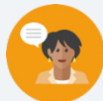
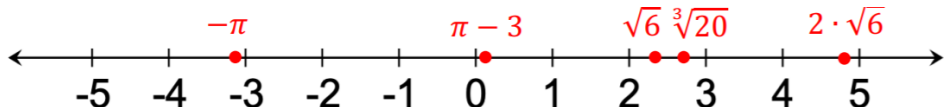
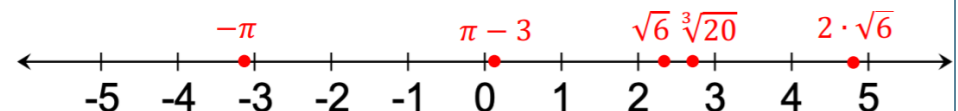


2.4 Approximating Irrational Numbers

Lesson Introduction

 Benchmarks	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line. MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers represented in various forms.			 Learning Targets	- I can identify whether a number is rational or irrational. - I can approximate the values of irrational numbers. - I can plot irrational numbers on a number line.
 Benchmark Coherence	Building On	Working Toward			
	N/A	MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system. MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.			
 Essential Questions	- How are square roots and cube roots similar and different? - How do you identify whether a number is rational or irrational? - How do you approximate the value of irrational numbers? - How do you plot irrational numbers on a number line?				
 Lesson Narrative	Students apply their knowledge of square roots to develop an understanding of cube roots. They then compare rational numbers and irrational numbers and identify each using definitions. Students approximate nonperfect square and cube roots and plot the approximated values on a number line. This work will prepare them to compare, order, and plot rational and irrational numbers on number lines in the next two lessons.				
 Component Summary	2.4.1 Warm-Up (~5 min.)	Complete a number pattern (<i>SE p. 80</i>)	2.4.4 Exploration (10 min.)	Estimate nonperfect square and cube roots using the nearest perfect squares and cubes (<i>SE p. 83</i>)	
	2.4.2 Collaboration (5-10 min.)	Summarize learning related to solving equations and evaluating square and cube roots (<i>SE p. 80</i>)	2.4.5 Guided Instruction (10 min.)	Approximate irrational numbers and plot them on a number line (<i>SE p. 85</i>)	
	2.4.3 Guided Instruction (5-10 min.)	Define and compare rational and irrational numbers (<i>SE p. 82</i>)	2.4.6 Guided Practice (5-10 min.)	Practice approximating irrational numbers and plotting them on a number line (<i>SE p. 86</i>)	
	2.4.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 86</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Irrational number - Principal square root		N/A		Students will need access to a calculator for 2.4.3 Guided Instruction, but then calculators should not be used for the remainder of this lesson or for the subsequent lessons.
	<u>Additional Terms:</u> N/A				

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse square roots with cube roots because they do not pay attention to the index. - Students may struggle approximating irrational square and cube roots to the nearest tenth. - Students may mistakenly think that pi equals 3.14 or $\frac{22}{7}$ rather than recognizing these are only approximations. - Students may mistakenly identify $\frac{\pi}{3}$ as rational because it is written as a ratio.	- Students know the definition of and have worked with rational numbers in the 6th grade accelerated course. - Remind students that when approximating the value of irrational numbers, first consider which whole numbers the nonperfect square or cube root falls in between. Then consider which whole number it is closer to. Finally, test a few decimals by squaring or cubing them to determine which is the best approximation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 2.4.2 Collaboration offers an opportunity for students to develop a deeper understanding of how to estimate principal square roots by engaging in rich mathematical conversations with their partners and then seeing and hearing from a variety of their classmates during the whole-class debrief. Before engaging in these discussions, prompt students to first attempt the questions independently.	MA.K12.MTR.3.1: Mathematical Fluency Students explore how to approximate nonperfect square and cube roots in 2.4.4 Exploration by applying their knowledge of perfect squares and cubes. Students begin developing fluency with these skills as practice exercises and discussions with their partners and the whole class, which will continue during subsequent lessons.
 Differentiate Instruction	In 2.4.4 Exploration, prompt students to generate a list of the perfect squares and a list of the perfect cubes to reference as a scaffold.	
	Consider creating a classroom number line where students can plot the square and cube roots during the debrief of 2.5.2 Collaboration, 2.4.5 Guided Instruction, and 2.4.6 Guided Practice. Physically plotting the points on a classroom number line and explaining their placement of the point provides kinesthetic, visual, and audio entry points.  Also, consider providing students with sentence stems to structure their responses in 2.4.7 Wrap-Up.	
Lesson Notes:		



2.4.1 Warm-Up: Number Pattern

Component Overview: Students identify a missing value in a pattern and explain their reasoning.



2.4.1 Warm-Up: Number Pattern

A pattern is shown with one value missing.

		47			
		25		22	
12		13		9	
2		10		3	6

1. Determine the value of the empty cell in the pattern.

47

2. Explain how you determined the missing value.

Answers will vary. For a diagonal pair, such as 12 and 10, the difference is 2, the value to the left of the lower number. This pattern holds for the diagram.



Discuss how the relationship between the values could be described as subtractive or additive. The subtractive description is provided in the answer key. The additive relationship is that the sum of any two adjacent numbers is the number between them in the next row up.



Encourage students to make their own puzzles that involve patterns. Have partners exchange patterns, find missing values, and explain their reasonings. Challenge students to incorporate perfect squares or roots into their puzzles.

2.4.2 Collaboration: Square Roots vs. Cube Roots

Component Overview: Students collaborate to solve questions related to what they have learned in Lessons 1-3 about square roots, cube roots, and solving equations involving perfect squares and perfect cubes.



2.4.2 Collaboration: Square Roots vs. Cube Roots

Work with your partner to complete the following.

- Damion knows that $8^2 = 64$ and $4^3 = 64$. He says this means the following equations are all true.

Equation A	$\sqrt{64} = 8$
Equation B	$\sqrt[3]{64} = 4$
Equation C	$\sqrt{-64} = -8$
Equation D	$\sqrt[3]{-64} = -4$

- Discuss with your partner whether or not you agree that all of the equations Damion wrote are true.
Answers will vary. We agree that Equations A, B, and D are correct but Equation C is incorrect. This is because you cannot find the square root of a negative number. Equation C is incorrect because the square root of -64 is a non-real solution.
- If Damion wrote any false equation(s), explain what makes them false.
If all of Damion's equations are true, explain how you know.
Answers will vary. Equation C is incorrect because you cannot find the square root of a negative number. Equation C is incorrect because the square root of -64 is a non-real number.



Some students may also identify Equation A as an incorrect answer because only one of the two solutions is listed. Consider using this as an opportunity to discuss why perfect squares have two roots but perfect cubes only have one. Relate this to the fact that negative radicands are possible for cube roots. Questions 2 illustrates this by having students expand out the multiplication.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

2.4.2 Collaboration: Square Roots vs. Cube Roots (continued)

2. Complete each statement.

a. $\sqrt[3]{-1}$ is ☐ equal to -1 ☐ not a real number because

- ☐ $-1 \times -1 \times -1 = -1$.
☐ _____ $\times$ _____ $= -1$.
☐ no real number multiplied by itself results in -1 .

b. $\sqrt{-1}$ is ☐ equal to _____ ☐ not a real number because

- ☐ _____ $\times$ _____ $\times$ _____ $= -1$.
☐ _____ $\times$ _____ $= -1$.
☐ no real number multiplied by itself results in -1 .

3. Work with your partner to complete the table by choosing a value of k based on the inequality. Then, determine if the result is real or non-real.

Description	Example	Real or Non-Real Result(s)
$\sqrt{k}$ where $k > 0$	Answers will vary. $\sqrt[3]{4} = \pm 2$; $\sqrt[3]{9} = \pm 3$	☐ real ☐ non-real
$\sqrt{k}$ where $k < 0$	Answers will vary. $\sqrt{-25} = \text{non-real}$; $\sqrt{-16} = \text{non-real}$	☐ real ☐ non-real
$\sqrt[3]{k}$ where $k > 0$	Answers will vary. $\sqrt[3]{64} = 4$, $\sqrt[3]{8} = 2$	☐ real ☐ non-real
$\sqrt[3]{k}$ where $k < 0$	Answers will vary. $\sqrt[3]{-64} = -4$ $\sqrt[3]{-8} = -2$	☐ real ☐ non-real

4. Write a short explanation of when a non-real result will occur.

Answers will vary. If you multiply two positive numbers the answer is positive, and when you multiply two negative numbers, the answer is also positive. Therefore, a non-real result will occur when you try to find the square root of a negative number because the result of squaring a real number is always positive.



Students may struggle to interpret the algebraic description in the table in question 3. Consider providing a written or verbal description and a few possible values students can use as examples. Modeling with a Think Aloud may help students understand the task expectations. Also, remind students that if no index is written, it is assumed to be 2.

2.4.3 Guided Instruction: Rational and Irrational Numbers

Component Overview: Students review the definitions of rational and irrational numbers. They revisit the concept of π , an example of an irrational number, from Grade 6 Accelerated. Students enter perfect and nonperfect square and cube roots in their calculators and determine whether they are rational or irrational based on the decimal expansions.



2.4.3 Guided Instruction: Rational and Irrational Numbers

The real number system includes both **rational** and **irrational numbers**.



A **rational number** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.



An **irrational number** is a real number that cannot be expressed as a ratio of two integers.

When expressed as decimals, irrational numbers do not terminate or repeat. Therefore, any decimal expansion is only an approximate value.

1. The relationship between the circumference and diameter of any circle is represented by π .

- a. Discuss with your partner what you remember about the value of π .

Answers will vary. Pi is the ratio of the circumference to the diameter of a circle. The circumference divided by the diameter results in pi every time. You can multiply a circle's diameter by pi to get its circumference. Pi is 3.14159... it goes on forever and never repeats, which means it is an irrational number. Sometimes we use 3.14 or $\frac{22}{7}$ to approximate the value of pi when solving problems.



Ask students to share their reasoning as they determine whether numbers are rational or irrational throughout the lesson, referring back to the definitions.



Question 1, part a is a discussion opportunity for the teacher to listen to determine students' prior knowledge, if any, about the value of π . Students will delve deeper into the meaning and value of π in Unit 8 when they explore circumference, area, and sectors of circles. For this unit, students only need to understand that π is an irrational number and what values are used to approximate it.

After listening to students' discussion, determine if there is any prior knowledge students have about π that could be shared with the group. If not, explain that π is a number representing the relationship between the circumference and area of a circle, a relationship that will be further explored in Unit 8. Then, prompt students to enter π in their calculator using the π button and/or search the value of π on google to show more places. Students can then discuss why π is an irrational number based on the definition of irrational numbers and what their calculator is showing.

2.4.3 Guided Instruction: Rational and Irrational Numbers (continued)

- b. Complete the statements.

The ratio of the circumference of a circle to its diameter is expressed as π . The values 3.14 and $\frac{22}{7}$

are ☐ equivalent to, or the exact value of π . The ☐ used to approximate the value of π . The

exact value of π ☐ can ☐ cannot be expressed as a fraction of two integers or a decimal. Therefore, π is

☐ a rational ☐ an irrational number.



Completion of these statements should be led by the teacher. The goal is for students to have the information needed to approximate values with pi while understanding that pi is an irrational number from the discussions in part a.

2. Enter each of the following values into a calculator.

$$\sqrt{25}$$

$$\sqrt{18}$$

$$\sqrt[3]{27}$$

$$\sqrt[3]{30}$$

- a. Circle the values that can be expressed as a ratio of two integers.
- b. Discuss with your partner how you would describe the values that are not circled.

Answers will vary. The square and cubed roots of the values not circled are not perfect squares or perfect cubes so they are irrational numbers. The square and cubed roots of the values not circled are non-perfect therefore they are irrational numbers.

- c. Complete the statements.

Square roots of perfect squares and cube roots of perfect cubes ☐ can ☐ cannot be expressed as a ratio of

two integers. Therefore, they are ☐ rational. ☐ irrational.

Square roots of non-perfect squares and cube roots of non-perfect cubes ☐ can ☐ cannot be expressed as a

ratio of two integers. Therefore, they are ☐ rational. ☐ irrational.

2.4.4 Exploration: Roots of Non-Perfect Squares and Non-Perfect Cubes

Component Overview: Students estimate the principal square root of nonperfect squares using the nearest perfect squares. Students are introduced to the fact that the square roots of nonperfect squares are irrational numbers. They then estimate the cube roots of nonperfect cubes using the nearest perfect cubes.

2.4.4 Exploration: Roots of Non-Perfect Squares and Non-Perfect Cubes

The decimal expansions of irrational numbers do not terminate nor repeat. Therefore, the values of **principal square root** of a non-perfect square can only be approximated.

The **principal square root** is the positive square root of a positive real number.

When evaluating the square root of a number, the **principal square root** is assumed when the radical symbol is used. A negative symbol, $-$, is placed in front of the radical to indicate the negative square root. Perfect squares can be used to estimate the value of the principal square root of non-perfect squares.

1. Work with your partner to complete the following to approximate the principal square root of $\sqrt{14}$.
 - a. Complete the inequality to show which two perfect squares 14 lies between.
 $9 < 14 < 16$
 - b. Use your answers to Part A to indicate which two square roots $\sqrt{14}$ lies between.
 $3 < \sqrt{14} < 4$
 - c. Discuss with your partner which square root the value of $\sqrt{14}$ is closer to.
 4
 - d. Plot $\sqrt{14}$ on the number line to indicate its approximate value based on your discussion.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group in 2.4.5 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



If students are stuck as they work with their partners through question 1, consider asking the following:

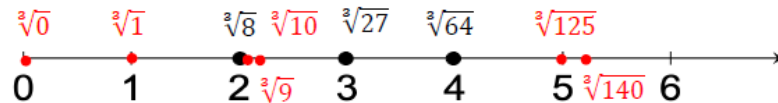
- *How did you determine which two perfect squares 14 lies between?*
- *What is the difference between 9 and 14? Between 14 and 16? How can this help you to determine which square root $\sqrt{14}$ is closest to?*
- *How will you show this on the number line?*

2.4.4 Exploration: Roots of Non-Perfect Squares and Non-Perfect Cubes (continued)

Similar to the square roots of non-perfect squares, the cube roots of some numbers are irrational.

The values of such cube roots can be approximated using the cube roots of perfect cubes.

2. A number line is shown with values indicated at specific points.



- a. Explain why $\sqrt[3]{8}$ is placed in the same location as 2 on the number line.

Answers will vary. It is placed there because 2 is the cubed root of 8. You multiply $2 \times 2 \times 2$ to get 8 therefore the cubed root of 8 is 2.

- b. Label the locations of the following cube roots on the number line.

$\sqrt[3]{1}$, $\sqrt[3]{125}$, and $\sqrt[3]{0}$

- c. Plot and label the approximate locations of the following cube roots on the number line.

$\sqrt[3]{9}$, $\sqrt[3]{10}$, and $\sqrt[3]{140}$

- d. Compare your answers to your partner's. Discuss any differences and make revisions to your work as needed.



Consider prompting students to generate a list of the perfect cubes from one to six to reference.



If students are stuck as they work with their partners through question 2, consider asking the following:

- How did you determine which two perfect cubes 9 lies between?
- What is the difference between 9 and 8? Between 9 and 27? How can this help you to determine which cube root $\sqrt[3]{9}$ is closest to?
- How can this be shown on the number line?

2.4.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line

Component Overview: Students learn to approximate $\sqrt{13}$ and expressions containing the $\sqrt{13}$ and locate the resulting approximations on number lines.

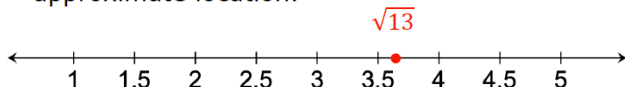


2.4.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line

The values of irrational numbers and expressions involving irrational numbers can be approximated.

1. Consider the expression $\sqrt{13}$.
 - a. Between which two whole numbers does $\sqrt{13}$ lie on a number line?
3 and 4
 - b. Which whole number is $\sqrt{13}$ is closer to?
4
 - c. Faith thinks the value of $\sqrt{13}$ will be close to 3.6 or 3.7. Discuss with your partner why Faith may have made these estimates.
Answers will vary. $3^2 = 9$ and $4^2 = 16$. 13 is a little more than halfway between the perfect squares 9 and 16. Faith estimates 3.6 or 3.7 because that is a little more than halfway between 3 and 4 on the number line.
 - d. Check Faith's estimates to determine if either estimate is close to the value of $\sqrt{13}$.
 $3.7 \cdot 3.7 = 13.69$ whereas $3.6 \cdot 3.6 = 12.96$ so $\sqrt{13}$ is closer to 3.6.
 - e. Based on your work in Part D, complete the statement.

The approximate value of $\sqrt{13}$, to the nearest tenth, is 3.6.
 - f. Plot and label $\sqrt{13}$ on the number line to indicate its approximate location.



Model squaring 3.6 and 3.7 to determine whether the decimal approximation for $\sqrt{13}$ is closer to 3.6 or 3.7. Explain that $(3.6)^2 = 12.96$ and $(3.7)^2 = 13.69$. Therefore, 3.6 is a closer approximation.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

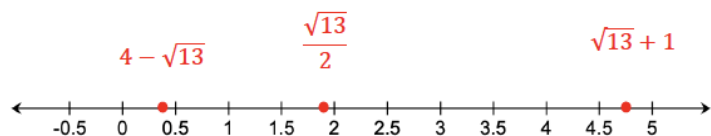
2.4.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line (continued)

2. Use the approximation of $\sqrt{13}$ from the previous question to complete the following.

- a. Approximate the value of each expression.

Expression	Approximate Value
$\sqrt{13} + 1$	4.6
$\frac{\sqrt{13}}{2}$	1.8
$4 - \sqrt{13}$	0.4

- b. Plot and label the approximate location of each expression from part a on the number line.



To approximate the value of expressions involving irrational numbers, first approximate the value of the irrational number. Then, use that approximation to perform any operations indicated.



While modeling how to approximate the value of each expression, consider using a Think Aloud, asking and answering the following:

- How did you approximate each expression?
- In each expression _____, which operation is performed first?



Use a Think Aloud to model how to plot and label each expression on the number line.

2.4.6 Guided Practice: Approximating Irrational Numbers and Locating Them on a Number Line

Component Overview: Students approximate the values of irrational numbers and plot their locations on a number line.

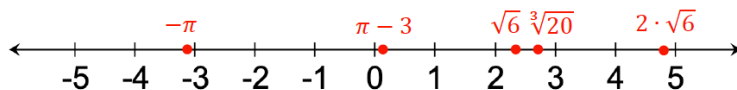


2.4.6 Guided Practice: Approximating Irrational Numbers and Locating Them on a Number Line

1. Approximate the value of each expression.

Expression	Approximate Value
$-\pi$	-3.1
$\sqrt{6}$	2.4
$\sqrt[3]{20}$	2.7
$2 \cdot \sqrt{6}$	4.8
$\pi - 3$	0.1

2. Plot and label the approximate location of each expression from the previous question on the number line.



Consider creating a classroom number line during the debrief of this activity. Physically plotting the points on a classroom number line and explaining their placement of the points provides kinesthetic, visual, and audio entry points.



If needed, consider asking the following scaffolding questions.

- How did you determine which whole numbers the approximation falls in between?
- How did you determine the decimal approximation of each expression?
- How can you check the accuracy of your approximations?

2.4.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



2.4.7 Wrap-Up: Cool Down

1. Explain the process for approximating an irrational number. Include each step in your process.

Answers will vary.